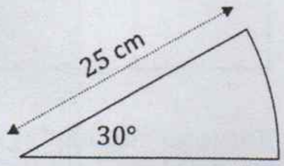
2nd Yr Maths Exam : November 2025 Assessment: Name: Oisin Argaud

85% $\frac{105}{123}$

Q1.

(a) A circular pizza of radius 25 cm is cut into equal-sized slices. Each slice is in the shape of a sector of angle 30°, as shown in the diagram.



(i) Write down the diameter of the pizza.

25×2

$\boxed{50\text{cm}}$

5

(ii) How many slices was the pizza cut into?

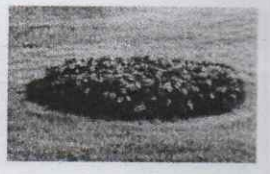
$360 \div 30$

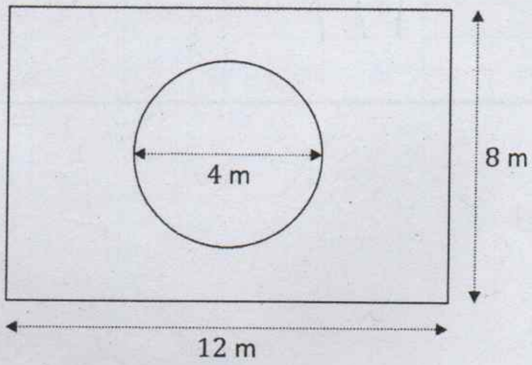
$\boxed{12 \text{ slices}}$

5

Q2.

Sarah has a circular flower bed of diameter 4 m in the centre of her garden.
The garden is in the shape of a rectangle 12 m long and 8 m wide, as shown in the diagram below.
The diagram is **not** to scale.





(a) Work out the area of the garden, in m².

12×8

$\boxed{96 m^2}$

$4 \div 2$
2

πr^2
$\pi(2)^2$
12.566370614

$96 - 12.566370614$
$83.433629386 m^2$

5

(b) Write down the radius of the flower bed. $4 \div 2$ 2 Radius = 2 m

5

20

(c) Work out the area of the flower bed.
Give your answer in m^2, correct to one decimal place.

$4 \div 2$
2

πr^2
$\pi(2)^2$
$\boxed{12.6\,m^2}$

10

(d) The remainder of the garden is made up of a grass lawn.
Work out the area of the grass lawn.
Give your answer in m^2, correct to one decimal place.

$4 \div 2$
2

πr^2
$\pi(2)^2$
12.6

8×12
96

$96 - 12.6$
$\boxed{83.4\,m^2}$

10

(e) Sarah wishes to plant the flower bed. She needs 14 plants to cover 1 m^2.
How many plants does she need to fill the flower bed?

πr^2
$\pi(2)^2$
$12.6\,m^2$

12.6×14
$\boxed{176.4 \text{ plants}}$

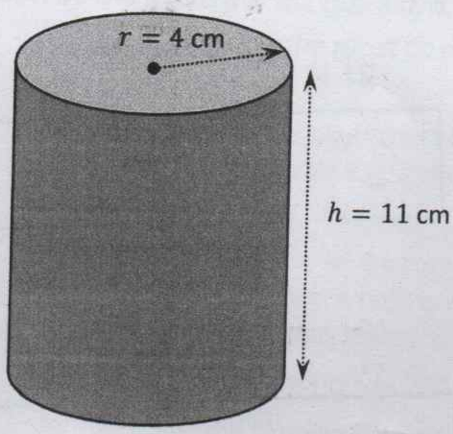
177

8/10

24/30

Q3.

Martina buys a carton of yoghurt. The carton is roughly in the shape of a cylinder.
It has the dimensions shown in the diagram below.



(a) Work out the **volume** of the carton in cm^3.
Give your answer in terms of π.

$V = \pi r^2 h$

$\pi(4)^2(11)$

$\boxed{176\pi \text{ cm}^3}$

10

Q4.

Solve the following questions on indices , write answer in index form (NOT an actual number value)

e.g. $3^2 \times 3^5 = 3^7$

i) $2^4 \times 2^3$

2^7 3

ii) $\frac{7^5}{7^2}$

7^3 3

iii) $(3^2)^3$

3^6 3

iv) $(5^3)^0$

5^0 3

v) $\frac{10^2 \times 10^3}{10^4}$

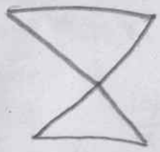
$\frac{10^5}{10^4}$ 10^1 3

vi) $\frac{x^3 \times x^2}{x^4}$

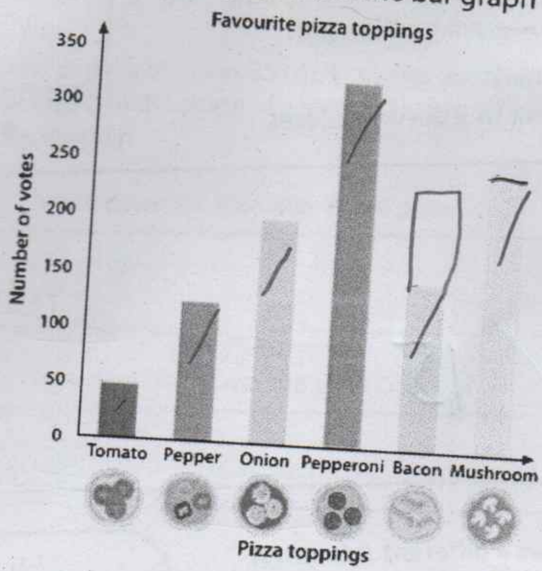
$\frac{x^5}{x^4}$ x^1 3

Q5.

18



Luigi's Pizzas took a survey about customers' favourite toppings and recorded the results in a bar graph. Use the bar graph below to answer the questions.



(i) Which is the most popular topping?

(ii) How many customers have chosen either tomato or pepper toppings?

(iii) If 75 additional customers prefer bacon, which one will be more popular, bacon or onion?

(iv) Which topping has 250 votes? Mushroom

(v) List the toppings in order from most popular to least popular.

(vi) How many customers were surveyed?

Most popular topping is Pepperoni

2

Number of customers whove have chosen either tomato or pepper toppings =
50+125 175 customers

2

If 75 additional customers prefer bacon, which one will be more popular, bacon or onion?
150+75=225 Bacon will be more popular

2

How many people were surveyed?
50+125+200+325+150+250 1100 people

2

List the toppings in order from the most popular to the least popular.
Pepperoni, Mushroom, Onion, Bacon, Pepper, Tomato

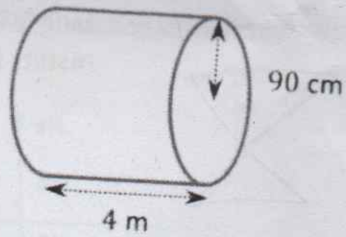
2

Q6

10/10

A tank for spreading fertiliser is in the shape of a cylinder.

The radius of the tank is 90 **centimetres** and its length is 4 metres, as shown in the diagram on the right.



Work out the **volume** of the tank.
Give your answer in m^3, correct to 2 decimal places.

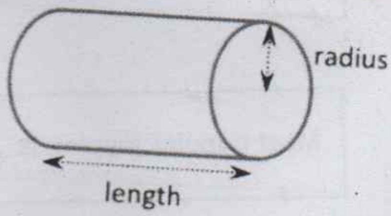
$\pi r^2 h$

$\pi (0.9)^2 (4)$

$\boxed{10.18\ m^3}$

✓ 15

This diagram on the right shows a different cylindrical tank, which has a volume of $\frac{81\pi}{125}$ m^3.



Its length is **three times** as long as its radius, as shown.

Work out the radius of this tank, in metres.

$\frac{81\pi}{125} = \pi r^2 h$ ✓

$0.8 = r \times h$

$0.8 \div 4 = 0.2$

$\boxed{0.2\ m = radius}$

4/20

$\frac{81}{125} = r^2 \times h$

$\sqrt{\frac{81}{125}} = \sqrt{r^2 \times h}$

$\frac{9}{11.2} = r \times h$

$\pi r^2 h = \frac{81\pi}{125}$

$\cancel{\pi} r^2 \cdot (3r) = \frac{81\cancel{\pi}}{125}$

$3r^3 = \frac{81}{125}$

$r^3 = 0.216$

$r = \sqrt[3]{0.216}$

$r = 0.6$